



ARBA MINCH UNIVERSITY
WATER TECHNOLOGY INSTITUTE

FACULTY OF WATER RESOURCES & IRRIGATION ENGINEERING

1. Discuss irrigation water application methods and criteria's to select the appropriate application methods.
2. What are the input parameters considering in design of irrigation canal?
3. List and discuss the drainage system components with the help of diagram.
4. Arba Minch demonstration farm has a gross area of 5000m². The dominant soil throughout the field is loam soil. The elevations are collected from 11 points. The grid

points are taken 25*25m. $S_x \text{ \& } S_y = \begin{cases} 0.4 & \text{If, calculated slope} > 0.4 \\ 0.2 & \text{If, calculated slope} < 0.2 \end{cases}$

Based on the surveyed elevation,

- a) Determine the centroid of the filed
- b) Determine the average elevation of the field
- c) Compute the slope of the plane of best fit
- d) Show the direction of drainage
- e) Compute the formation levels, cuts and fills

1.78	1.65	1.52	
1.79	1.67	1.53	1.36
1.80	1.70	1.55	1.41

5. Design an irrigation channel to carry a discharge of 5m³/s in an alluvial soil by Kennedy's theory. The side slopes of channel is 0.5:1 and assume critical velocity ratio (m=0.8), the channel bed slop is 1 in 5000. Use the Chezy's equation for estimation of actual velocity with value of C is 30. Take first trial depth of flow as 1m.
6. Design an irrigation channel to carry a discharge of 5m³/s in an alluvial soil by Kennedy's Theory. The side slopes of channel is 0.5:1 and assume critical velocity ratio (m=0.8), the channel bed slop is 1 in 5000. Use the Chezy's equation for

estimation of actual velocity with value of C is 30. Take first trial depth of flow as 1m. (7 mark)

7. What does it mean land leveling and list the benefits of land leveling?
8. Discuss briefly the water application methods and the selecting criteria's.
9. What will occur
 - a) If the mean velocity is greater than the permissible velocity?
 - b) If the mean velocity is less than the permissible velocity?
10. Assume you are a drainage engineer, if a certain irrigation field is found in between depression (Undulating topography), which types of drainage systems is recommend? Why?
11. The design of masonry lined irrigation canal is proposed to carry canal capacity of $9.5\text{m}^3/\text{s}$. The canal have longitudinal slope of 0.015% with permissible velocity of 0.71m/s and manning coefficient of 0.016. Determine the canal parameter by considering side slope of canal 1:1 and finally draw its cross- section.
12. Design a trapezoidal open canal for 3000ha area. If the duty at the head is 10lps/ha. The channel is to be laid at a slope of 1m/km. The critical velocity ratio for the soil is Use Kutters rugosity coefficient as 0.023.
13. Design a triangular drainage ditch to drain the excess water accumulated on the soil surface of 700ha irrigated area. The longitudinal slope of the field on which the drainage ditch running is 1m/Km. The side slope, and manning's coefficient of the canal is 1:1 and 0.04 respectively. The mean annual rainfall in the area is 980mm. (Hint: Use Hudson empirical methods)